

Pattern-aware Illicit Account Detection based on User Behavior Sequences

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Abstract

Illicit activities such as financial fraud and fake promotions are increasingly prevalent on social applications. The diverse behavioral structures of different types of illicit accounts pose difficulties in designing unified detection strategies. Unlike existing studies that focus on binary detection and often rely on user static profiles, we introduce a novel task: illicit account detection based solely on user behavior sequences. This task presents two key challenges: 1) illicit accounts often mimic benign users by performing normal-looking behavior subsequences, and 2) behaviors with the same action (e.g., add-friend, initiate-transaction) can serve different purposes, such as illicit or benign. To address these challenges, we propose a Pattern-aware Illicit Accounts Detection (PIAD) framework that consists of three components: 1) a dual-perspective pattern mining module that extracts category-specific self- and interaction-behavior patterns from behavior sequences to capture distinct behavioral regularities across different user types; 2) a contextualized action semantic encoding algorithm that aligns action codings with contextual dependencies among behaviors within user sequences to capture variations in purposes when behaviors with the same actions occur under different contexts; and 3) a pattern-aware fusion model that integrates the mined patterns with the context and interaction in behavior sequences to learn discriminative representations for detection. Extensive experiments on real-world datasets demonstrate that PIAD consistently outperforms state-of-the-art baselines with an average 7.51% improvement on F1 score.

CCS Concepts

• **Computing methodologies** → *Machine learning*; • **Security and privacy** → *Intrusion/anomaly detection and malware mitigation*.

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Illicit Account Detection, Behavior Sequence Modeling, Pattern Mining and Fusion

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1 Introduction

Social applications, characterized by massive user participation and rapid information dissemination [11, 18], have increasingly become hubs for illicit activities, with applications like Weixin and Facebook frequently targeted and exploited by cybercriminals [15, 32]. These malicious activities, including financial fraud [13, 41] and fake marketing schemes [1, 30], are often orchestrated through social interactions, posing severe threats to application security and account safety. Moreover, illicit accounts exhibit behavioral diversity [15, 19], resulting in difficulties in applying uniform countermeasures. Therefore, illicit account detection that identifies the specific behavioral strategies of different illicit types is essential for enabling governance policies and risk-aware mitigation [9, 25].

Recently, a few works [5, 36] focus on the illicit account detection task. Illicit accounts are often characterized by suspicious static profiles and distinctive regularities in their behavior sequences [15, 19]. However, existing approaches predominantly rely on user static profiles [5, 36], which raises privacy concerns and limits their applicability in practical scenarios where access to personal data is restricted or legally prohibited. In addition, most existing efforts remain limited to binary detection, only aiming to distinguish illicit accounts from benign ones, while overlooking the detection of fine-grained illicit types. These limitations call for studies that focus on the fine-grained detection of illicit accounts without relying on privacy-sensitive information.

In this work, we advocate the use of user behavior sequences as the foundation for illicit account detection. Unlike static profiles, behavior sequences contain only action traces without sensitive

content or identifiable attributes. The actions in these sequences are drawn from a predefined action set (e.g., add-friend, initiate-transaction, etc.), and each behavior represents a concrete occurrence of one such action at a specific time. Through the actions and context, these sequences embed the patterns of different types of users. For example, financial fraud accounts frequently exhibit rapid friend additions followed by transaction links, whereas fake marketing accounts tend to send repeated promotional links [42]. Motivated by this, we formulate the fine-grained detection task solely based on user behavior sequences.

There are two key challenges to this detection problem: 1) Illicit accounts often mimic benign users by performing normal-looking behavior subsequences [24]. However, existing detection methods lack mechanisms to model the discriminative behavior structures between different types of users, as they focus on simplified interaction information, such as binary user interactions [24] or the statistical information of interactions [32]. 2) The same behavior can serve different purposes, such as illicit or benign. For example, in financial fraud, a transaction-type behavior immediately following an add-friend behavior often indicates scamming, while the same behavior in the behavior sequence of a benign user may occur within daily chatting [37]. Lack of modeling context makes it difficult for models to differentiate the intents behind actions. However, most existing methods [24, 32] oversimplify behavioral data, overlooking such semantic and dependency nuances.

To address the aforementioned challenges, we propose the Pattern-aware Illicit Accounts Detection (PIAD) framework with three modules: a dual-perspective pattern mining (DPM) algorithm, a contextualized action semantic encoding (CASE) algorithm, and a pattern-aware information fusion model. To emphasize the distinct behavioral regularities of different user types, the DPM algorithm is designed to extract self- and interaction-behavior patterns that are unique to each user type. Then, the CASE algorithm merges contextual semantic differences into action coding by clustering actions and modeling their context dependencies via Markov transition matrices [23]. It aligns differences between action codings with contextual dependencies among behaviors, assisting in the capture of variations in purposes when behaviors with the same actions occur under different contexts. Finally, to learn discriminative representations for each user, the extracted patterns and encoded actions from these two algorithms are used to build the pattern-aware information fusion model that integrates a pattern-masked supervision-enhanced BERT (PS-BERT) and a pattern-aligned discriminative GNN (PD-GNN). PS-BERT pre-trains user embeddings from their behavior sequences, which are then fed into PD-GNN to incorporate structural information for final detection. The contributions are summarized as follows:

- This work studies a novel problem of behavior-level illicit account detection and presents PIAD, a framework integrating semantic-context encoding, dual-perspective pattern mining, and pattern-aware information fusion.
- A dual-perspective pattern mining algorithm is developed to extract category-specific self- and interaction-behavior patterns, capturing distinct behavioral regularities of different user types.

- A contextualized action semantic encoding algorithm aligns differences between action codings and contextual dependencies among behaviors, helping disambiguate identical actions with different purposes.
- A pattern-aware fusion model is designed to integrate mined patterns with contextual and interaction structures of behavior sequences, learning discriminative representations for detection.
- Extensive experiments on two real-world datasets show the effectiveness of PIAD, with up to a 7.51% improvement in F1 over state-of-the-art methods.

2 Related Works

2.1 Illicit Account Detection

Illicit account detection in social networks has become increasingly critical with the rise of large-scale financial fraud [13, 41] and fake marketing schemes [1, 30]. Early approaches rely on both user profile features and interactions [13, 20]. Farrugia et al. [6] demonstrate that XGBoost trained on account-derived features can effectively detect illicit accounts, while Tam et al. apply GCN-based graph representation learning on e-payment networks, leveraging edge-level embeddings for suspicious account detection. However, leveraging this information poses privacy risks and limits the applicability of these approaches.

Thus, recent studies have shifted toward modeling user interactions [4, 10]. Hu et al. [10] introduce BERT4ETH, a pre-trained Transformer that encodes transactions as sequences to capture multi-hop dependencies and mitigate data heterogeneity for phishing detection. Ding et al. [4] propose DIAM, a multigraph neural network with discrepancy-aware attention to model transaction sequences and emphasize differences between benign and illicit nodes. However, these methods only partially model interaction information: BERT4ETH does not explicitly represent interaction structure, while DIAM simplify diverse action types into coarse statistics. Such abstraction limits their ability to capture semantic variations and increases vulnerability when illicit accounts mimic benign behavior patterns.

In contrast, our task explicitly models interaction graphs with multi-typed user actions, enabling fine-grained behavioral representation and more robust discrimination between benign and malicious accounts.

2.2 Classification based on Behavior Sequence

User behavior sequences are central to social network analysis, driving tasks such as illicit account detection [4, 21] and recommender systems [8, 38]. Many methods exploit their sequential structure, applying RNNs [2] and Transformers [39] to embed timestamped actions as categorical tokens. Recent work further enhances temporal modeling of sequences. For instance, CBD [7] employs a content-aware balanced decoder with spectrum interaction modulation, while MPTSNet [17] decomposes sequences into multi-scale periodic segments to capture intra- and inter-period dependencies. Moreover, sequential pattern mining [28, 40] is studied to explicitly extract patterns from sequences. For instance, SPADE [40] mines frequent subsequences via vertical data format. Yet these methods focus on sequential patterns and ignore interaction structures.

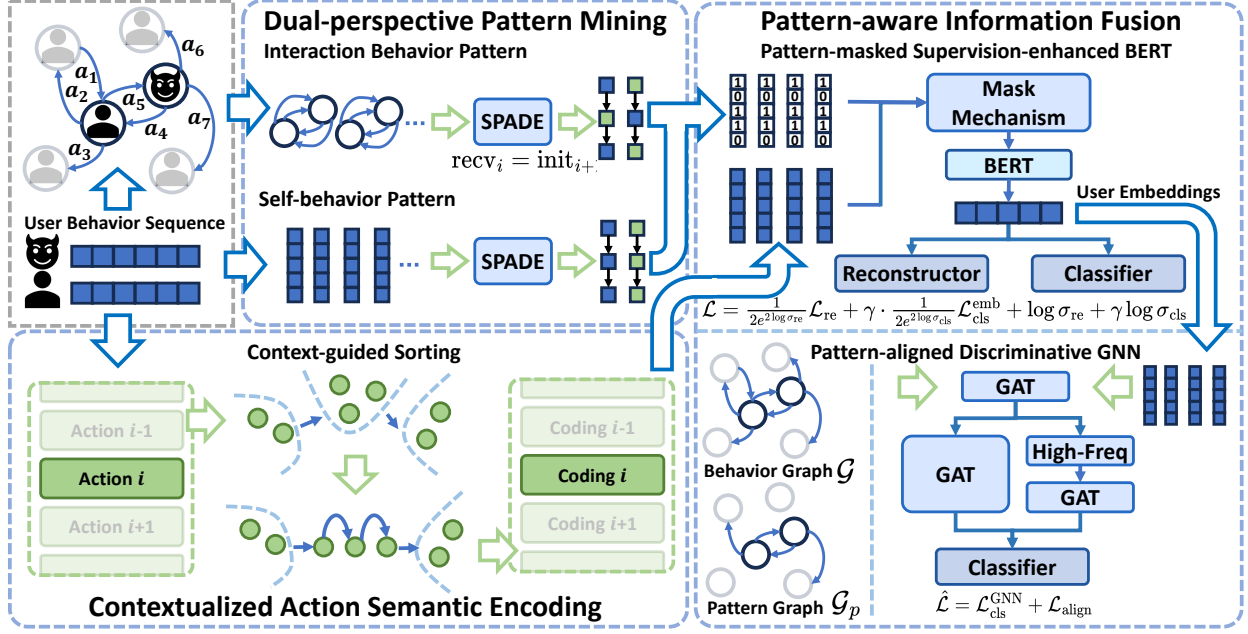


Figure 1: Overview of PIAD. Based on user behavior sequences, where behavior interaction tuples are able to be transformed into a behavior graph, PIAD first mines self- and interaction-level patterns and encodes actions with contextual semantics, and then integrates them into a pattern-aware fusion model combining a pattern-masked supervision-enhanced BERT for user embedding generation and a pattern-aligned discriminative GNN for detection.

More recently, Graph Neural Networks (GNNs) [22, 34, 35] have emerged as a powerful paradigm for user behavior modeling. Based on the message aggregation mechanism, GNNs encode structural dependencies between users. For instance, spatiotemporal GNNs [31] capture both temporal regularities and interaction structures in user activities. GSAT [16] introduces stochastic attention under an information bottleneck objective to filter task-irrelevant signals, highlighting the need for structure-aware feature learning.

However, existing approaches are insufficient for illicit account detection based on user behavior sequences because they address only isolated aspects of the problem. Motivated by these limitations, we propose a systematic framework that jointly models contextual semantic and structural information and integrates pattern-aware feature alignment, providing a comprehensive solution for illicit account detection.

3 Problem Definition

Let $\mathcal{U} = \{u_1, u_2, \dots, u_N\}$ denote the set of users in the dataset. Each user $u \in \mathcal{U}$ is associated with a sequence of behavioral interactions, denoted by $\mathcal{S}^u = \{(s_i, r_i, a_i, t_i)\}_{i=1}^{n^u}$, where each interaction tuple (s_i, r_i, a_i, t_i) represents an action initiated by user s_i and received by user r_i , with action type $a_i \in \mathcal{A}$ (e.g., adding friend, creating chatroom) and timestamp $t_i \in \mathbb{R}^+$. For every interaction related to user u , the user must appear as either the sender or the receiver: $\forall (s_i, r_i, a_i, t_i) \in \mathcal{S}^u, s_i = u \text{ or } r_i = u$.

Given the behavior sequence $\mathcal{S}^u$ of a user $u \in \mathcal{U}$, we consider two detection tasks:

1) Illicit Accounts Detection: The first task is to determine whether the user is illicit. Formally, the detection model is defined as:

$$M_{\text{det}} : \mathcal{S}^u \rightarrow \{0, 1\} \quad (1)$$

where $M_{\text{det}}(\mathcal{S}^u) = 1$ indicates that user u is classified as illicit, and $M_{\text{det}}(\mathcal{S}^u) = 0$ indicates a benign user.

2) Illicit Type Detection: The second task is to classify the user into fine-grained illicit categories. Specifically, let $\mathcal{Y} = \{0, 1, \dots, C\}$ be the label space, where 0 represents a benign user and 1 through C correspond to different illicit activity types. The classification model is defined as:

$$M_{\text{type}} : \mathcal{S}^u \rightarrow \mathcal{Y} \quad (2)$$

where $M_{\text{type}}(\mathcal{S}^u) = y$ indicates that user u belongs to the y .

4 Methodology

We propose the Pattern-aware Illicit Accounts Detection (PIAD) framework, whose overview is shown in Fig. 1. PIAD comprises three components: 1) a dual-perspective pattern mining algorithm extracting self- and interaction-behavior patterns, 2) a contextualized action semantic encoding algorithm capturing semantic proximity and contextual relevance, and 3) a pattern-aware fusion model combining contextual semantics and structural information via a supervision-enhanced BERT and a discriminative GNN.

Given user behavior sequences, PIAD first performs pattern mining and action encoding. The extracted patterns and encoded behavior sequences are then fed into a pattern-aware information

fusion model. This model first trains a pattern-masked supervision-enhanced BERT to generate user embeddings, which are then combined with a behavior graph built from the sequences and refined and classified by a pattern-aligned GNN.

4.1 Dual-Perspective Pattern Mining Algorithm

Illicit accounts often mimic benign users by performing normal-looking behavior subsequences, blurring the decision boundaries and posing challenges for account detection. To address this challenge, we introduce a pattern mining algorithm to extract category-specific behavioral patterns from two perspectives, including self-behavior patterns derived from user behavior sequences initiated by the same user, and interaction-behavior patterns extracted from sequences of user-to-user interactions.

4.1.1 Pattern Mining via SPADE Algorithm. To capture patterns in sequences, we utilize the SPADE algorithm [40], a widely used pattern mining method, which is a vertical mining algorithm that decomposes the pattern growth process into equivalence classes based on shared prefixes.

In detail, SPADE leverages a vertical data format, where each action is linked to the list of users and positions in which it appears. Given the behavior sequence S^u of each user u , an inverted index $I(a)$ is constructed for each action a , defined as $I(a) = \{(u, p) \mid S^u[p] = a\}$, where (u, p) indicates that action a occurs at position p in the behavior sequence of the user u .

Then, SPADE enumerates all users and positions where the pattern occurs in order. A k -length pattern $P = \langle a_1, \dots, a_k \rangle$ is associated with an occurrence list:

$$I(P) = \{(u, p_1, \dots, p_k) \mid S^u[p_i] = a_i, p_1 < \dots < p_k\} \quad (3)$$

Subsequently, SPADE recursively extends patterns by joining occurrence lists of patterns with single actions to generate longer candidates. Given a $(k-1)$ -length pattern $P_1 = \langle a_1, \dots, a_{k-1} \rangle$ and an action a_k , the new candidate pattern is $P = P_1 \circ a_k$, and the occurrence list is updated as:

$$I(P) = \{(u, p_1, \dots, p_k) \mid (u, p_1, \dots, p_{k-1}) \in I(P_1)\} \quad (4)$$

where $(u, p_k) \in I(a_k)$ and $p_{k-1} < p_k$. Moreover, to control the probability of occurrence of patterns, a candidate pattern P is retained if its user-level support, denoted as $\text{supp}(P)$, meets the minimum threshold σ :

$$\text{supp}(P) = |\{u \mid \exists (u, \cdot) \in I(P)\}| \geq \sigma \quad (5)$$

Finally, patterns that pass the support check are recursively extended until no further patterns can be generated.

Dual-Perspective Pattern Mining Algorithm. Based on SPADE, we conduct category-wise mining, extracting patterns separately for each user category c from all users $u^c \in \mathcal{U}^c$. For a given c , patterns are derived from two perspectives: 1) Self-behavior patterns, capturing users' internal routines from sequences containing only actions initiated by the same user $u^c \in \mathcal{U}^c$. 2) Interaction-behavior patterns, modeling dyadic turn-based dynamics from interleaved sequences between each user pair (u^c, v) with $u^c \in \mathcal{U}^c$.

To obtain self-behavior sequences, we filter S^{u^c} to retain only actions where u^c is the initiator:

$$S_{\text{self}}^{u^c} = \{a \in S^{u^c} \mid \text{init}(a) = u^c\} \quad (6)$$

The SPADE-based mining process is applied to $S_{\text{self}}^{u^c}$ for all $u^c \in \mathcal{U}^c$, via the occurrence list extension strategy, yielding the self-behavior patterns denoted as $\mathcal{P}_{\text{self}}^c$.

For interaction-behavior patterns, we partition S^{u^c} by interaction partners. For each $v \neq u^c$ such that v co-occurs with u in any action, we construct:

$$S_{\text{inter}}^{u^c, v} = \{a \in S^{u^c} \mid \{\text{init}(a), \text{recv}(a)\} = \{u^c, v\}\} \quad (7)$$

where S^{u^c} denotes the full interaction history involving u^c , including both actions initiated by u^c and actions where u^c is the receiver, ensuring that $S_{\text{inter}}^{u^c, v}$ captures the complete bidirectional dynamics for the user pair (u^c, v) . Each action a is represented as $a = (\text{act}, \text{init}, \text{recv})$ to preserve initiator–receiver metadata.

To mine turn-taking patterns which reflect the regularity of interaction between users, we impose the initiator–receiver alternation constraint that requires the receiver of one action to become the initiator of the next in any pattern $P = (a_1, a_2, \dots, a_i, \dots, a_k)$, formulated as $\text{recv}[a_i] = \text{init}[a_{i+1}], \forall i \in [1, k-1]$.

The interaction-behavior pattern set for category c , which is denoted as $\mathcal{P}_{\text{inter}}^c$, is obtained via performing SPADE on each $S_{\text{inter}}^{u^c, v}$ within the category c , retaining a candidate pattern only if the alternation constraint is satisfied.

Category-Exclusive Pattern Filtering. After mining, we apply a de-duplication step across categories to retain only unique patterns for each category:

$$\mathcal{P}^c = \left\{ P^c \mid P^c \in (\mathcal{P}_{\text{self}}^c \cup \mathcal{P}_{\text{inter}}^c) \wedge \forall c' \neq c, \text{supp}^{c'}(P^c) = 0 \right\} \quad (8)$$

where $\text{supp}^{c'}(P^c) = |\{u^{c'} \in \mathcal{U}^{c'} \mid \exists (u^{c'}, \cdot) \in I(P^c)\}|$ denotes the user-level support of pattern P^c in category c' .

To incorporate both self-behavior and interaction-behavior patterns into representation learning, we construct a binary mask $\mathbf{m}^{u^c} \in \{0, 1\}^{|S^{u^c}|}$ for each user $u^c \in \mathcal{U}^c$. For each position t in S^{u^c} , the $m_t^{u^c} \in \mathbf{m}^{u^c}$ is defined as follows:

$$m_t^{u^c} = \begin{cases} 1, & \text{if } t \text{ is positions of any } P \in \mathcal{P}^c \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

The pattern mask $\mathbf{m}^{u^c}$ marks discriminative positions where frequent patterns occur, enabling pattern-aware masking and embedding during model training.

4.2 Contextualized Action Semantic Encoding

To capture the contextual variations of actions in user behavior sequences, we propose a contextualized action semantic encoding algorithm. The motivation is ranking actions with similar semantics based on their Markov transition probabilities to reflect their context relevance, and then initializing action coding accordingly.

4.2.1 Semantic Clustering via Keyword Grouping. Actions that appear in user behavior sequences are represented by textual descriptions containing key semantic terms. To capture coarse-grained semantic proximity, we cluster actions based on shared keywords in their literal descriptions.

Formally, let $\mathcal{K}(a_i)$ denote the set of keywords extracted from action $a_i \in \mathcal{A}$. Actions sharing identical keyword sets are grouped

into the same cluster $C_j = \{a_i \mid \mathcal{K}(a_i) = \kappa_j\}$, where κ_j is the j -th unique keyword set.

4.2.2 Context-guided Sorting via Markov Transition Matrices. To reflect the context dependencies present in behavior sequences, we model the context proximity between actions using Markov transition matrices [23].

For each user group, we construct two transition probability matrices: 1) $M_g \in \mathbb{R}^{m \times m}$: the transition matrix between each semantic cluster, where m is the number of clusters. 2) $M_l \in \mathbb{R}^{N \times N}$: the transition matrix between individual action items. Specifically, the cluster-level transition probability is defined as $M_g(i, j) = \frac{\text{count}(C_i \rightarrow C_j)}{\sum_{j'} \text{count}(C_i \rightarrow C_{j'})}$ and similarly, $M_l(i, j)$ is computed by normalizing the one-step transitions between action items $M_l(i, j) = \frac{\text{count}(a_i \rightarrow a_j)}{\sum_{j'} \text{count}(a_i \rightarrow a_{j'})}$.

Based on M_g , we extract cluster transition chains to reveal high-level context structures. Starting from each cluster C_i , we recursively follow the most probable dependence $C_i \rightarrow C_j$ until reaching a terminal cluster or forming a loop.

In the presence of a cycle, we select the cluster with the highest occurrence probability $P(C_v) = \sum_{a \in C_v} P(a)$ as the entry point, then unwrap the cycle into a directed chain. All resulting chains are ranked based on the occurrence probabilities of their starting clusters, yielding an ordered cluster sequence that reflects global behavioral context.

Then, the actions in each cluster are ordered to capture local context. For each action $a_i \in C_v$, we compute its cumulative transition score based on its context relations with actions in the adjacent clusters C_{v-1} and C_{v+1} :

$$s(a_i) = \frac{\sum_{a_j \in C_{v-1}} M_l(j, i)}{\sum_{a_k \in C_{v+1}} M_l(i, k)} \quad (10)$$

which quantifies the extent to which a_i is contextually influenced by the previous cluster and influences the next cluster. Actions in C_v are then ranked in descending order of $s(a_i)$.

By concatenating the sorted actions from all clusters according to the ordered cluster sequence, we encode the actions sequentially to obtain the coding vector $\mathcal{O} = [o_1, o_2, \dots, o_N]$, where o_i denotes the code assigned to action a_i . The behavior sequence of each user $u \in \mathcal{U}$ is then updated as $\hat{S}^u = (s_i, r_i, o_i, t_i)_{i=1}^{n^u}$.

4.3 Pattern-Aware Information Fusion Model

Leveraging the mined dual-perspective patterns and encoded actions, we propose a pattern-aware fusion model that integrates contextual semantics and structural information into unified, discriminative user embeddings.

The model operates in a two-stage workflow: 1) A pattern-masked supervision-enhanced BERT (PS-BERT) is first trained to generate user embeddings by reconstructing mined patterns and aligning them with user types. 2) A pattern-aligned discriminative GNN (PD-GNN) refines embeddings generated in 1) by fusing interaction structures of behavior sequences and inducing their features to remain consistent with the features from pattern behaviors, enabling the final detection.

4.3.1 Pattern-masked Supervision-enhanced BERT. The first component of our fusion model leverages a modified BERT encoder to

capture semantic and context dependencies within user behavior sequences. Unlike standard BERT with random masking, our pattern-masked supervision-enhanced BERT additionally masks out entire mined behavior patterns, forcing the model to reconstruct them from the surrounding context and learn coherent action chains.

For each user u , we extract the initiator-side behavior sequence $\hat{o}^u = \{o \in \hat{S}^u \mid \text{init}(o) = u\}$ and standardize it to a fixed length w . Following the BERT input format, a [CLS] token is prepended and a [SEP] token is appended. If the raw sequence length $l < w$, [SEP] is inserted after the last action, and the remainder is padded with [PAD]; if $l \geq w$, the sequence is truncated to w actions and terminated with [SEP]. Thus, the final input is $\mathbf{x}^u = ([\text{CLS}], \hat{o}_1^u, \dots, \hat{o}_w^u, [\text{SEP}], [\text{PAD}], \dots)$, where each $\hat{o}_i \in \hat{o}^u$, and the sequence has a total length of $w + 2$.

In parallel, we apply the pattern mask $\mathbf{m}^u$ from Eq. 9, where $m_i^u = 1$ indicates $\hat{o}_i^u$ is part of a mined pattern. Masking is restricted to $\hat{o}_i^u$ in $\mathbf{z}^u$, while special tokens ([CLS], [SEP], [PAD]) are kept unchanged. The masked sequence $\tilde{\mathbf{x}}^u$ is defined as:

$$\tilde{x}_t^u = \begin{cases} [\text{MASK}], & x_t^u \in \{\hat{o}_i^u \mid m_i^u = 1\}, \\ [\text{MASK}] \text{ with prob. } p, & x_t^u \in \{\hat{o}_i^u \mid m_i^u = 0\}, \\ x_t^u, & \text{otherwise.} \end{cases} \quad (11)$$

Then, the masked sequence is fed into a BERT encoder [3] to get the hidden representation $\mathbf{H}^u = \text{BERT}(\tilde{\mathbf{x}}^u)$, where $\mathbf{H}^u = (\mathbf{h}_{[\text{CLS}]}^u, \mathbf{h}_1^u, \dots, \mathbf{h}_w^u, \mathbf{h}_{[\text{SEP}]}^u, \mathbf{h}_{[\text{PAD}]}^u, \dots)$, and $\mathbf{h}_t^u \in \mathbb{R}^d$ is the representation for x_t^u .

For each masked position $t \in \mathcal{M}$, a reconstructor predicts the original token as $\hat{y}_t^{re} = g_{re}(\mathbf{h}_t^u)$.

To connect reconstructed patterns with user type, the hidden representation of [CLS] is selected as the embedding of user u and fed into a classification head $\hat{y}^{cls} = g_{cls}(\mathbf{h}_{[\text{CLS}]}^u)$.

Thus, the reconstruction and classification losses are:

$$\mathcal{L}_{re} = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \text{CrossEntropy}(\hat{y}_t^{re}, x_t^u) \quad (12)$$

$$\mathcal{L}_{cls}^{\text{emb}} = \text{CrossEntropy}(\hat{y}^{cls}, y^u) \quad (13)$$

where $\mathcal{M} = \{t \mid \tilde{x}_t^u = [\text{MASK}]\}$ and y^u is the ground-truth label of the user u .

To adaptively balance the two objectives, we follow the uncertainty-weighted multi-task formulation [12] with a decayed classification factor γ :

$$\mathcal{L} = \frac{1}{2\sigma_{re}^2} \mathcal{L}_{re} + \log \sigma_{re} + \gamma \left(\frac{1}{2\sigma_{cls}^2} \mathcal{L}_{cls}^{\text{emb}} + \log \sigma_{cls} \right) \quad (14)$$

where σ_{re} and σ_{cls} are learnable parameters, and γ decays exponentially with the training epoch to gradually reduce the weight of $\mathcal{L}_{cls}^{\text{emb}}$. Through this joint optimization, the encoder learns to reconstruct semantically coherent behavior patterns while aligning user-level embeddings with class semantics.

Finally, after training PS-BERT via $\mathcal{L}$, the embedding $\mathbf{h}_{[\text{CLS}]}^u$ for each $u \in \mathcal{U}$ is generated from PS-BERT and input into the following PD-GNN.

4.3.2 Pattern-aligned Discriminative GNN. While the PS-BERT captures contextual semantics within user sequences, user behaviors

also form structural regularities through the behavior graph. To integrate structural information to refine user embeddings, we design a pattern-aligned discriminative GNN.

First, we construct a behavior graph $\mathcal{G} = (\mathcal{U}, \mathcal{E}, \mathcal{Z})$ from behavioral tuples $(s_i, r_i, o_i, t_i) \in \mathcal{S}^u$ for each user $u \in \mathcal{U}$, where each directed edge (s_i, r_i) with an action o_i as attribute. The initial node feature vector $\mathbf{z}^u \in \mathcal{Z}$ is the BERT-derived sequence embeddings $\mathbf{h}_{[\text{CLS}]}^u$. We further derive a pattern graph $\mathcal{G}_p = (\mathcal{U}, \mathcal{E}_p)$ containing only behaviors participating in mined patterns $\mathcal{P}$.

Considering that users with different types often interact with each other, we design two GAT-based channels to capture both global consistency and cross-category variations:

$$\mathbf{Z}_{\text{sim}} = \text{GAT}(\mathcal{Z}, \mathcal{E}), \quad \mathbf{Z}_{\text{diff}} = \text{GAT}_{\text{high}}(\mathbf{H}_{\text{high}}\mathcal{Z}, \mathcal{E}) \quad (15)$$

where $\mathbf{H}_{\text{high}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ is a high-pass filter with adjacency matrix $\mathbf{A}$ and degree matrix $\mathbf{D}$ derived from $\mathcal{E}$. $\mathbf{Z}_{\text{sim}} \in \mathbb{R}^{|\mathcal{U}| \times d}$ captures global trends, while $\mathbf{Z}_{\text{diff}} \in \mathbb{R}^{|\mathcal{U}| \times d}$ highlight cross-category variations.

The two representation channels are combined through a learnable gating mechanism. For each user $u \in \mathcal{U}$, the gate value ϕ^u is computed as

$$\phi^u = \sigma(\mathbf{W}_g [\mathbf{z}_{\text{sim}}^u \parallel \mathbf{z}_{\text{diff}}^u]) \quad (16)$$

where $[\cdot \parallel \cdot]$ denotes vector concatenation, $\mathbf{W}_g \in \mathbb{R}^{d \times 2d}$ is a learnable weight matrix, and $\sigma(\cdot)$ is the sigmoid function.

The final fused representation is then obtained as

$$\hat{\mathbf{z}}^u = \phi^u \odot \mathbf{z}_{\text{sim}}^u + (1 - \phi^u) \odot \mathbf{z}_{\text{diff}}^u \quad (17)$$

where $\odot$ denotes element-wise multiplication.

Moreover, a residual connection with the original BERT embedding is added before classification:

$$\hat{\mathbf{y}}^u = \text{Softmax}(\mathbf{W}_2 \cdot \text{ELU}(\mathbf{W}_1 \hat{\mathbf{z}}^u + \mathbf{W}_r \mathbf{z}^u)) \quad (18)$$

where $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{W}_r \in \mathbb{R}^{d \times d}$ are trainable parameters.

The supervised classification loss is defined as follows:

$$\mathcal{L}_{\text{cls}}^{\text{GNN}} = - \sum_{u \in \mathcal{U}} y^u \log \hat{\mathbf{y}}^u \quad (19)$$

where y^u is the ground-truth label.

To enhance the sensitivity of the model to the mined patterns, a pattern alignment loss is designed to enforce consistency of the hidden representation $\tilde{\mathbf{z}}^u$ and $\tilde{\mathbf{z}}_p^u$ that are extracted from $\mathcal{G}$ and $\mathcal{G}_p$, respectively:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \|\tilde{\mathbf{z}}^u - \tilde{\mathbf{z}}_p^u\|_2^2 \quad (20)$$

Finally, the total loss of the PS-GNN is defined as $\hat{\mathcal{L}} = \mathcal{L}_{\text{cls}}^{\text{GNN}} + \mathcal{L}_{\text{align}}$. The PD-GNN is trained with $\hat{\mathcal{L}}$ to fuse structural information into the user embeddings generated by PS-BERT, which are subsequently utilized to perform detection.

5 Experiment

5.1 Experimental Setups

5.1.1 Datasets. We evaluate our approach on four real-world datasets collected from the WeChat platform, spanning two types of illicit accounts (Fraud and Non-fraud) and two data collection durations, 1 day (1d) and 5 days (5d). Due to platform privacy constraints, user

profile attributes are not accessible, and all datasets are constructed solely from user behavior sequences.

Each dataset consists of fine-grained temporal sequences of user actions, capturing interaction dynamics between initiators and targets. User-level labels are assigned to initiators, enabling both binary illicit account detection and fine-grained classification of illicit user types. The two duration variants are independently sampled rather than temporally nested, ensuring that user sets are disjoint across datasets. Basic dataset statistics are reported in Table 4.

The Fraud datasets focus on deceptive behaviors involving financial intent, while the Non-fraud datasets cover policy-violating behaviors without direct financial fraud. Detailed descriptions of illicit account categories, behavior sequences, and the action taxonomy are provided in Appendix A.

5.1.2 Baselines. We compare PIAD with nine baselines spanning illicit account detection (BERT4ETH [10] and DIAM [4]), sequence classification (TS-GAC [33], CBD [7], and MPTSNet [17]), and graph classification (GCN [14], GAT [29], EI-CDA [26], and GSAT [16]). Our task focuses on illicit account detection over interaction graphs, where edges correspond to multi-typed user actions and the interaction structure is explicitly modeled. Most prior fraud detection methods [10] operate on individual behavior sequences and do not capture such interaction structures. Detailed descriptions and distinctions are provided in Appendix B.

5.1.3 Implementation Details. For pattern mining, the threshold σ in the dual-perspective pattern mining algorithm is fixed at 0.02 for illicit account detection and 0.06 for illicit type detection on the Fraud dataset, and 0.5 for all tasks on the Non-Fraud dataset. For the pattern-aware supervision-enhanced BERT, the learning rate is set to 1×10^{-4} with 8 transformer layers and a maximum input sequence length of $w = 200$. For the pattern-aligned discriminative GNN, the learning rate is 0.01. Both modules are trained for 30 epochs with a batch size of 64, a dropout rate of 0.3, and a weight decay of 1×10^{-5} . For baseline methods, we follow the authors' original implementations. To ensure consistency, the maximum sequence length is unified to 200 across all models, and the random seed is fixed at 0 to guarantee reproducibility.

We evaluate model performance using four standard metrics: Accuracy (Acc.), Precision (Pre.), Recall (Rec.), and Macro F1-score (F1). Macro F1 is adopted as the primary metric since it averages per-class F1-scores without weighting, providing robustness to class imbalance and better reflecting performance on minority illicit categories. All experiments are run for five iterations, and the average results are reported. All experiments are performed on a single NVIDIA H20 GPU.

5.2 Performance Analysis

Table 1 reports illicit account detection results, and Table 2 shows illicit type detection. PIAD consistently achieves the best performance, with average F1 gains over 2.53% on account detection and about 12.12% on type detection compared to the strongest baseline.

Overall, the results show that illicit type detection is significantly more challenging than illicit account detection. Fine-grained identification requires modeling subtle distinctions between different

Table 1: Results for illicit account detection. The best results are highlighted in bold, while sub-optimal results are underlined.

Datasets	1d Fraud				5d Fraud				1d Non-fraud				5d Non-fraud			
Methods	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
GCN	65.11	65.30	65.05	64.94	67.33	68.49	67.42	66.88	80.06	80.48	79.99	79.92	72.27	72.27	72.26	72.26
GAT	60.57	51.08	60.42	53.49	63.00	58.22	63.07	59.62	81.67	81.77	81.74	81.67	71.70	72.47	71.72	71.46
EI-CDA	60.08	60.71	58.24	56.65	52.03	51.52	50.74	44.14	63.05	65.67	61.12	59.18	55.17	56.29	55.70	54.32
GSAT	68.07	70.51	67.47	66.62	67.49	68.71	67.63	67.06	80.28	82.52	80.26	79.91	71.30	75.58	70.87	69.74
TS-GAC	67.27	67.73	67.31	67.09	70.22	70.22	70.22	70.22	93.55	94.63	93.04	93.42	80.77	80.76	80.71	80.73
CBD	68.75	69.38	68.84	68.56	69.14	69.14	69.13	69.13	82.42	82.38	82.43	82.40	76.29	76.39	76.15	76.18
MPTSNet	69.26	69.46	69.02	68.99	73.63	74.07	73.71	73.55	89.15	89.38	89.14	89.13	87.37	87.37	87.37	87.37
BERT4ETH	<u>76.33</u>	<u>76.66</u>	<u>76.08</u>	<u>76.12</u>	<u>81.99</u>	<u>82.05</u>	<u>81.98</u>	<u>81.97</u>	<u>97.44</u>	<u>97.52</u>	<u>97.50</u>	<u>97.44</u>	<u>92.30</u>	<u>92.37</u>	<u>92.33</u>	<u>92.30</u>
DIAM	74.24	75.14	69.33	71.98	77.24	76.04	81.04	78.44	86.75	90.29	82.63	86.25	89.86	87.66	92.18	89.84
PIAD	80.33	80.42	80.40	80.32	84.82	84.87	84.82	84.81	99.94	99.94	99.93	99.94	92.88	92.89	92.89	92.88

Table 2: Results for illicit type detection. The best results are highlighted in bold, while sub-optimal results are underlined.

Datasets	1d Fraud				5d Fraud				1d Non-fraud				5d Non-fraud			
Methods	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
GCN	53.66	8.34	8.95	7.53	52.73	7.89	8.75	7.80	60.14	25.02	28.52	24.75	56.18	27.92	25.56	24.89
GAT	54.16	9.16	8.98	7.73	51.71	10.12	8.00	7.03	59.87	33.91	28.81	25.23	55.71	28.73	25.10	23.82
EI-CDA	52.69	3.36	6.11	4.34	50.22	8.09	7.92	6.91	56.01	19.94	18.63	15.07	52.69	17.32	17.09	13.52
GSAT	54.68	<u>22.60</u>	9.37	9.51	51.51	16.42	8.32	7.89	60.34	34.71	21.11	20.62	54.21	30.79	18.18	17.46
TS-GAC	52.69	9.01	10.12	9.11	53.20	13.14	9.54	8.97	<u>69.79</u>	<u>39.90</u>	<u>37.76</u>	<u>37.51</u>	59.26	32.82	25.20	26.47
CBD	52.86	12.87	12.34	<u>11.63</u>	52.97	16.62	9.62	9.36	62.89	36.87	33.23	33.74	59.15	35.99	26.33	28.51
MPTSNet	51.70	13.29	9.17	8.47	53.51	12.23	10.38	10.01	66.28	27.32	24.73	24.80	61.87	42.20	28.59	28.43
BERT4ETH	<u>58.57</u>	10.27	13.13	11.49	<u>60.89</u>	19.79	16.89	17.37	62.52	34.65	34.51	32.33	<u>74.53</u>	50.66	<u>47.39</u>	48.16
DIAM	54.53	8.17	10.43	8.39	57.28	17.66	13.70	13.41	59.36	29.42	25.11	24.54	66.44	40.44	36.83	36.84
PIAD	60.90	28.74	24.36	24.50	69.17	51.08	41.11	45.18	78.02	44.97	42.84	42.45	76.10	53.88	51.97	52.51

illicit categories, making it inherently more complex. When comparing the Fraud and Nonfraud datasets, the latter consistently achieves higher scores. Pattern mining analysis indicates that Non-Fraud behaviors tend to exhibit more unique interaction motifs and a higher pattern repetition, which facilitates their separation and leads to improved detection performance.

Regarding the baselines, models that encode edge sequences to generate embeddings, such as BERT4ETH and DIAM, generally outperform other approaches. BERT4ETH ranks just behind PIAD in most account detection settings, benefiting from BERT’s capacity to capture action semantics. TS-GAC achieves its best result on Nonfraud 1d, likely due to its shallow Transformer design matching the simpler patterns in this dataset. However, in datasets where patterns are less explicit, TS-GAC exhibits no significant advantage.

Overall, PIAD demonstrates clear superiority, combining context semantic modeling with pattern-aligned representations to deliver robust improvements across both account and type detection tasks.

5.3 Ablation Study Analysis

We evaluate the contribution of each component of PIAD through ablation experiments, as shown in Table 3. The w/o Pattern-aware variant removes the pattern-guided supervision in the information fusion module, w/o Action Encoding encodes actions using only

semantic similarity without contextual relevance, and w/o GNN discards the discriminative GNN in favor of using only BERT-based embeddings. Removing any of these modules consistently degrades performance, confirming their effectiveness. For instance, on the Fraud 1d dataset for the illicit account detection, the F1 score of PIAD is 80.32, while w/o Pattern-aware, w/o Action Encoding, and w/o GNN drop to 79.24, 80.28, and 78.31.

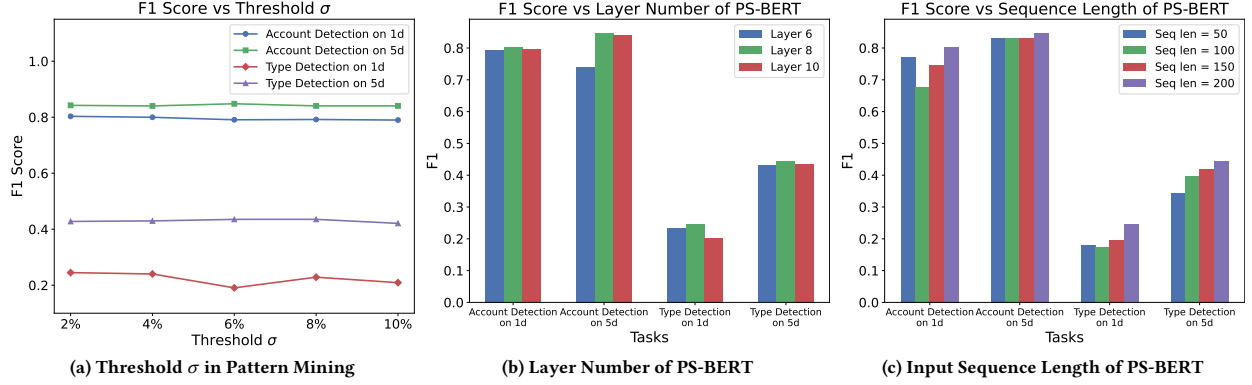
We also examine two replacement settings. PIAD with BERT substitutes the pattern-masked supervision-enhanced BERT with a standard BERT, while PIAD with GNN replaces the pattern-aligned discriminative GNN with a simple GAT. Both lead to significant drops; on the Fraud 1d dataset, the F1 score decreases to 59.82 and 71.04 respectively. These results highlight that pattern-masked supervision enables BERT to learn more discriminative representations, while the pattern-aligned GNN effectively captures heterophilous user interactions by emphasizing high-frequency signals that simple aggregation fails to model.

5.4 Hyperparameter Analysis

We evaluate the sensitivity of PIAD to three key hyperparameters on the Fraud datasets for both illicit account detection and illicit type detection: 1) the threshold σ in pattern mining, 2) the layer number of BERT, and 3) the input sequence length of BERT.

Table 3: Ablation Study Results of PIAD on Fraud Datasets.

Datasets	Illicit Account Detection								Illicit Type Detection							
	1d Fraud				5d Fraud				1d Fraud				5d Fraud			
Methods	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
PIAD	80.33	80.42	80.40	80.32	84.82	84.87	84.82	84.81	60.90	28.74	24.36	24.50	69.76	51.24	41.78	45.18
w/o Pattern-aware	79.35	79.54	79.20	79.24	84.27	84.33	84.28	84.27	59.55	24.81	22.93	22.13	69.08	50.21	40.89	44.09
w/o CASE	80.29	80.38	80.35	80.28	84.14	84.19	84.15	84.14	59.27	25.76	20.66	21.39	68.28	45.32	34.85	37.93
w/o PD-GNN	78.33	78.43	78.37	78.31	82.61	82.66	82.62	82.60	<u>60.82</u>	<u>31.63</u>	22.05	22.91	<u>69.50</u>	<u>52.28</u>	38.24	42.35
PIAD with BERT	64.61	64.11	65.18	59.82	81.26	81.31	81.27	81.26	58.73	30.86	16.83	18.30	66.41	45.50	28.94	33.35
PIAD with GNN	72.69	72.49	72.90	71.04	<u>84.48</u>	84.51	<u>84.48</u>	<u>84.47</u>	58.37	27.17	<u>23.57</u>	<u>23.58</u>	69.26	52.54	39.93	43.89

**Figure 2: Effect of the threshold σ in pattern mining, the layer number and the input sequence length of pattern-masked supervision-enhanced BERT on the Fraud datasets for both illicit account detection and type detection.**

Effect of Threshold σ in Pattern Mining. Fig. 2a shows the impact of the repeat probability of mined patterns. For illicit account detection, the F1 score on the 1d dataset peaks at 2% (80.32) and declines to 78.99 at 10%, while the performance on the 5d dataset remains stable near 84 with the best value at 6% (84.81). For illicit type detection, the performance on the 1d dataset drops sharply beyond 4%, falling from 24.50 to 19.06 at 6%, whereas the performance on the 5d dataset reaches its highest score at 6% (43.52). These results suggest that moderate repeat probabilities mitigate overfitting and prevent redundant patterns from misleading the model.

Effect of PS-BERT's Layer Number. We analyze the effect of the PS-BERT's layer number. As shown in Fig. 2b, increasing layers from 6 to 8 consistently improves the F1 scores. For illicit account detection, the performance on the 5d dataset rises from 73.97 to 84.81, and for illicit type detection, the performance on the 1d dataset improves from 23.29 to 24.50. However, moving to 10 layers reduces performance, with the 1d dataset dropping to 20.13 and the 5d dataset slightly to 83.98. This indicates that 8 layers provide the best trade-off; deeper networks introduce more parameters, which can harm learning when data or features are insufficient.

Impact of Maximum Sequence Length. Fig. 2c demonstrates that extending the input length of the PS-BERT enhances performance by capturing richer temporal dependencies. On the illicit account

detection, performance on the 1d dataset increases from 77.29 (50) to 80.32 (200), while on the illicit type detection, the performance on the 1d dataset grows from 17.99 to 24.50. For the 5d dataset on the illicit type detection, the F1 score rises markedly from 34.27 to 45.18. Longer sequences preserve more historical behaviors, enabling the model to detect both illicit accounts and types more accurately.

6 Conclusion

In this work, we address the problem of behavior-level illicit account detection and propose a effective framework named PIAD, which integrates semantic-context encoding, dual-perspective pattern mining, and pattern-aware fusion to jointly capture behavior semantics and discriminative patterns. Experiments on large-scale real-world datasets indicate that PIAD achieves consistent improvements up to 7.51% on the F1 score over state-of-the-art baselines in both illicit account detection and illicit type detection. By capturing semantic variations and discriminative patterns, PIAD provides a principled solution for modeling complex illicit behaviors and offers practical value for securing social applications.

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References

- [1] Malak Aljabri and Rami Mustafa A Mohammad. 2023. Click fraud detection for online advertising using machine learning. *Egyptian Informatics Journal* 24, 2 (2023), 341–350.
- [2] Don Dennis, Durmus Alp Emre Acar, Vikram Mandikal, Vinu Sankar Sadasivan, Venkatesh Saligrama, Harsha Vardhan Simhadri, and Prateek Jain. 2019. Shallow rnn: accurate time-series classification on resource constrained devices. *Advances in neural information processing systems* 32 (2019).
- [3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 4171–4186.
- [4] Zhihao Ding, Jieming Shi, Qing Li, and Jiannong Cao. 2024. Effective Illicit Account Detection on Large Cryptocurrency MultiGraphs. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management* (Boise, ID, USA) (CIKM '24). Association for Computing Machinery, New York, NY, USA, 457–466. doi:10.1145/3627673.3679707
- [5] Youssef Elmougy and Ling Liu. 2023. Demystifying fraudulent transactions and illicit nodes in the bitcoin network for financial forensics. In *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining*. 3979–3990.
- [6] Steven Farrugia, Joshua Ellul, and George Azzopardi. 2020. Detection of illicit accounts over the Ethereum blockchain. *Expert Systems with Applications* 150 (2020), 113318.
- [7] Yudong Han, Haocong Wang, Yupeng Hu, Yongshun Gong, Xuemeng Song, and Weili Guan. 2025. Content-aware Balanced Spectrum Encoding in Masked Modeling for Time Series Classification. *Proceedings of the AAAI Conference on Artificial Intelligence* 39 (04 2025), 17059–17067. doi:10.1609/aaai.v39i16.33875
- [8] Zhicheng He, Weiwen Liu, Wei Guo, Jiarui Qin, Yingxue Zhang, Yaochen Hu, and Ruiming Tang. 2023. A survey on user behavior modeling in recommender systems. *arXiv preprint arXiv:2302.11087* (2023).
- [9] Chuanbo Hu, Bin Liu, Yanfang Ye, and Xin Li. 2023. Fine-grained classification of drug trafficking based on Instagram hashtags. *Decision Support Systems* 165 (2023), 113896.
- [10] Sihao Hu, Zhen Zhang, Bingqiao Luo, Shengliang Lu, Bingsheng He, and Ling Liu. 2023. BERT4ETH: A Pre-trained Transformer for Ethereum Fraud Detection. In *Proceedings of the ACM Web Conference 2023* (Austin, TX, USA) (WWW '23). Association for Computing Machinery, New York, NY, USA, 2189–2197. doi:10.1145/3543507.3583345
- [11] Xingting Ju, Raquel Chocarro, and Oscar Martín Martín. 2021. Value creation in mobile social media: A systematic review and agenda for future research. *Baltic Journal of Management* 16, 5 (2021), 745–764.
- [12] Alex Kendall, Yarin Gal, and Roberto Cipolla. 2018. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 7482–7491.
- [13] Samira Khodabandehlou and Alireza Golpayegani. 2024. FiFraud: unsupervised financial fraud detection in dynamic graph streams. *ACM Transactions on Knowledge Discovery from Data* 18, 5 (2024), 1–29.
- [14] TN Kipf. 2016. Semi-Supervised Classification with Graph Convolutional Networks. *arXiv preprint arXiv:1609.02907* (2016).
- [15] Meng Li. 2022. A survey on ethereum illicit detection. In *International Conference on Artificial Intelligence and Security*. Springer, 222–232.
- [16] Siqi Miao, Mia Liu, and Pan Li. 2022. Interpretable and Generalizable Graph Learning via Stochastic Attention Mechanism. In *Proceedings of the 39th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 162)*, Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato (Eds.). PMLR, 15524–15543. <https://proceedings.mlr.press/v162/miao22a.html>
- [17] Yang Mu and Xiao Zhu. 2025. MPTNet: Integrating Multiscale Periodic Local Patterns and Global Dependencies for Multivariate Time Series Classification. *Proceedings of the AAAI Conference on Artificial Intelligence* 39 (04 2025), 19572–19580. doi:10.1609/aaai.v39i18.34155
- [18] Hua Pang, Yuxin Qiao, and Kaige Zhang. 2024. Modeling pathway linking mobile social media intensity to attitude towards electronic word-of-mouth and engagement: the significant role of social trust and perceived homophily. *Technological Forecasting and Social Change* 198 (2024), 123023.
- [19] Anna Popik-Mazur. 2025. A systematic literature review of illicit financial flows and money laundering: Current state of research and estimation methods. *Journal of Economics and Management* 47, 1 (2025), 257–298.
- [20] Sidharth Samanta, Bhabendu Kumar Mohanta, Siba Prasad Pati, and Debasish Jena. 2019. A framework to build user profile on cryptocurrency data for detection of money laundering activities. In *2019 International Conference on Information Technology (ICIT)*. IEEE, 425–429.
- [21] Dhara Shah, TG Harrison, Christopher B Freas, David Maimon, and Robert W Harrison. 2020. Illicit activity detection in large-scale dark and opaque web social networks. In *2020 IEEE international conference on big data (Big Data)*. IEEE, 4341–4350.
- [22] Kartik Sharma, Yeon-Chang Lee, Sivagami Nambi, Aditya Saliyan, Shlok Shah, Sang-Wook Kim, and Srijan Kumar. 2024. A survey of graph neural networks for social recommender systems. *Comput. Surveys* 56, 10 (2024), 1–34.
- [23] Chris Sherlaw-Johnson, Steve Gallivan, and Jim Burridge. 1995. Estimating a Markov transition matrix from observational data. *Journal of the Operational Research Society* 46, 3 (1995), 405–410.
- [24] Jie Song, Sijia Zhang, Pengyi Zhang, Junghoon Park, Yu Gu, and Ge Yu. 2024. Illicit Social Accounts? Anti-Money Laundering for Transactional Blockchains. *IEEE Transactions on Information Forensics and Security* (2024).
- [25] Jie Song, Sijia Zhang, Pengyi Zhang, Junghoon Park, Yu Gu, and Ge Yu. 2025. Illicit Social Accounts? Anti-Money Laundering for Transactional Blockchains. *IEEE Transactions on Information Forensics and Security* 20 (2025), 391–404. doi:10.1109/TIFS.2024.3518068
- [26] Yongduo Sui, Shuyao Wang, Jie Sun, Zhiyuan Liu, Qing Cui, Longfei Li, Jun Zhou, Xiang Wang, and Xiangnan He. 2025. A Simple Data Augmentation for Graph Classification: A Perspective of Equivariance and Invariance. *ACM Trans. Knowl. Discov. Data* 19, 2, Article 51 (Feb. 2025), 24 pages. doi:10.1145/3706062
- [27] Da Sun Handason Tam, Wing Cheong Lau, Bin Hu, Qiu Fang Ying, Dah Ming Chiu, and Hong Liu. 2019. Identifying Illicit Accounts in Large Scale E-payment Networks—A Graph Representation Learning Approach. *arXiv preprint arXiv:1906.05546* (2019).
- [28] Esther Ulitzsch, Qiwei He, and Steffi Pohl. 2022. Using sequence mining techniques for understanding incorrect behavioral patterns on interactive tasks. *Journal of Educational and Behavioral Statistics* 47, 1 (2022), 3–35.
- [29] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. Graph Attention Networks. *arXiv:1710.10903* [stat.ML]. <https://arxiv.org/abs/1710.10903>
- [30] Ranjeet Vishwakarma and Rajesh Dhakad. 2024. Online advertising and fraud click in online advertisement: A survey. *International Journal of Computer Applications* 186, 1 (2024), 0975–8887.
- [31] Daheng Wang, Meng Jiang, Munira Syed, Oliver Conway, Vishal Juneja, Sriram Subramanian, and Nitesh V Chawla. 2020. Calendar graph neural networks for modeling time structures in spatiotemporal user behaviors. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining*. 2581–2589.
- [32] Hongyu Wang, Ying Li, Ronghong Huang, and Xianghang Mi. 2025. Detecting and Understanding the Promotion of Illicit Goods and Services on Twitter. In *Proceedings of the ACM on Web Conference 2025*. 3389–3404.
- [33] Yucheng Wang, Yuecong Xu, Jianfei Yang, Min Wu, Xiaoli Li, Lihua Xie, and Zhenghua Chen. 2024. Graph-aware contrasting for multivariate time-series classification. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence (AAAI'24/IAAI'24/EAAI'24)*. AAAI Press, Article 1753, 10 pages. doi:10.1609/aaai.v38i14.29501
- [34] Lingfei Wu, Peng Cui, Jian Pei, Liang Zhao, and Xiaojie Guo. 2022. Graph neural networks: foundation, frontiers and applications. In *Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining*. 4840–4841.
- [35] Zonghan Wu, Shirui Pan, Fengwen Chen, Guodong Long, Chengqi Zhang, and Philip S Yu. 2020. A comprehensive survey on graph neural networks. *IEEE transactions on neural networks and learning systems* 32, 1 (2020), 4–24.
- [36] Zequan Xu, Lianyuan Li, Hui Li, Qihang Sun, Shaofeng Hu, and Rongrong Ji. 2023. Self-supervised graph representation learning for black market account detection. In *Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining*. 330–338.
- [37] Zequan Xu, Qihang Sun, Shaofeng Hu, Jieming Shi, and Hui Li. 2024. Crowdsourcing Fraud Detection Over Heterogeneous Temporal MMMA Graph. In *International Conference on Database Systems for Advanced Applications*. Springer, 327–339.
- [38] Enming Yuan, Wei Guo, Zhicheng He, Huifeng Guo, Chengkai Liu, and Ruiming Tang. 2022. Multi-behavior sequential transformer recommender. In *Proceedings of the 45th international ACM SIGIR conference on research and development in information retrieval*. 1642–1652.
- [39] Yuan Yuan and Lei Lin. 2020. Self-supervised pretraining of transformers for satellite image time series classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 14 (2020), 474–487.
- [40] Mohammed J Zaki. 2001. SPADE: An efficient algorithm for mining frequent sequences. *Machine learning* 42, 1 (2001), 31–60.
- [41] Ge Zhang, Zhao Li, Jiaming Huang, Jia Wu, Chuan Zhou, Jian Yang, and Jianliang Gao. 2022. eFraudcom: An e-commerce fraud detection system via competitive graph neural networks. *ACM Transactions on Information Systems (TOIS)* 40, 3 (2022), 1–29.
- [42] Yongchun Zhu, Dongbo Xi, Bowen Song, Fuzhen Zhuang, Shuai Chen, Xi Gu, and Qing He. 2020. Modeling users' behavior sequences with hierarchical explainable network for cross-domain fraud detection. In *Proceedings of the web conference 2020*. 928–938.

A Datasets

Illicit Account Categories. The Fraud datasets include multiple types of financial deception, such as romance scams, fake part-time job scams, and investment fraud. The Non-fraud datasets consist of users engaging in policy-violating behaviors, including malicious marketing, abusive automation, dissemination of pornographic content, and gambling promotion. Overall, the Fraud datasets are annotated with 16 labels (1 benign and 15 illicit types), while the Non-fraud datasets contain 10 labels (1 benign and 9 illicit types).

Behavior Sequences Structure. Each dataset comprises fine grained user behavior sequences. Each behavior sequence record includes the initiating user, the target user, a timestamp, and a detailed action descriptor. User-level labels indicate the behavioral role of the initiator rather than individual actions.

Action Taxonomy. User actions are organized into a hierarchical taxonomy. High-level categories include account registration, device login, friend interaction, content sharing, and other social behaviors. Each category is further subdivided to capture behavioral nuances and interaction patterns.

Sampling and Annotation. The 1d and 5d datasets are independently sampled rather than constructed via temporal extension. As a result, user sets are strictly disjoint across datasets, preventing information leakage between different time-span settings.

Table 4: Statistics of the four datasets covering two types of illicit accounts. The 1-day (1d) and 5-day (5d) datasets are independently sampled and contain disjoint user sets.

Statistics	Fraud		Non-fraud	
	1d	5d	1d	5d
# Users	57,583	474,676	53,290	631,403
# Behaviors	445,887	5,415,373	595,017	5,411,773
# Action types	97	103	102	108
# Avg. behaviors	178.00	279.73	349.60	239.80
# Illicit users	1,212	9,794	850	11,048
# Benign users	1,293	9,565	852	11,520
# Classes	16	16	10	10

B Baselines

This section provides detailed descriptions of the baselines used in our experiments and clarifies their relationship to our problem setting. Our task targets illicit account detection over interaction graphs, where nodes represent users, edges correspond to multi-typed user actions, and the underlying interaction structure is explicitly modeled.

Illicit Account Detection. DIAM [4] is the closest baseline to our setting, modeling transaction networks as multigraphs and using attention to highlight discrepancies between benign and illicit nodes. However, it simplifies the rich semantic information of edges, encoding them mainly through sequential representations and coarse aggregation, which limits its ability to distinguish subtle variations in user actions. BERT4ETH [10] treats illicit account detection as a sequence modeling problem, employing a pre-trained Transformer with masked token prediction and sequence deduplication. While

effective at capturing multi-hop dependencies, it does not fully exploit the interaction structure of user actions.

Sequence Classification. We include three sequence-based models to examine the limitations of approaches that ignore interaction structure. TS-GAC [33] enforces spatial consistency in sequences through graph-based augmentations and contrastive learning. CBD [7] integrates a content-aware decoder with spectrum modulation and energy rebalance to enhance feature diversity. MPTNet [17] decomposes behavior sequences into multi-scale segments and applies local-global extractors to capture intra- and inter-sequence patterns.

Graph Classification. We further compare with classical and recent graph classification methods that operate on static graph structures. GCN [14] learns node representations via spectral convolution and neighborhood aggregation, while GAT [29] applies multi-head attention for adaptive neighbor weighting. EI-CDA [26] mixes stable substructures and extracts environment-level features, and GSAT [16] leverages stochastic attention under an information bottleneck to suppress task-irrelevant correlations.

Overall, these baselines allow us to systematically evaluate the impact of modeling choices that omit either interaction structure or multi-typed user actions, thereby highlighting the necessity of jointly modeling both aspects in PIAD.

C Complexity Analysis

This section provides a complete analysis of the computational cost of the three major components of our framework, including the Dual-Perspective Pattern Mining (DPM), the Contextualized Action Semantic Encoding (CASE), and the Pattern-Aware Information Fusion Model.

C.1 Dual-Perspective Pattern Mining Algorithm.

SPADE mines frequent subsequences in the vertical occurrence list space [40]. All user sequences are scanned once to construct occurrence lists, incurring a preprocessing cost of $O(|\mathcal{U}| \cdot \bar{S})$, where $|\mathcal{U}|$ is the number of users and $\bar{S}$ is the average length of the user behavior sequences.

Subsequently, SPADE extends each frequent pattern by adding a new action and constructing the corresponding occurrence list. This extension is obtained by scanning the existing occurrence list of the pattern and checking, for each sequence identifier, whether the new event occurs in a valid subsequent position. The cost of such an extension is dominated by traversing the occurrence list and its position sets, and under a uniform approximation is $O(\bar{S})$ per extension.

Let N_{pat} denote the number of patterns that survive the minimum support threshold σ . Although the theoretical worst case yields $N_{\text{pat}} = O(2^{\bar{S}})$, real-world behavioral data are typically sparse and heavily pruned by the support threshold σ . Thus, the practical average-case complexity of SPADE can be written as

$$T_{\text{SPADE}} = O(|\mathcal{U}| \cdot \bar{S} + N_{\text{pat}} \cdot \bar{S}). \quad (21)$$

The space complexity follows the same reasoning. Storing all occurrence lists requires maintaining, for each frequent pattern, up

to $O(\bar{S})$ position entries across user sequences, yielding an average-case space consumption of

$$S_{\text{SPADE}} = O(N_{\text{pat}} \cdot |\mathcal{U}| \cdot \bar{S}). \quad (22)$$

Since DPM performs SPADE mining independently under the two complementary perspectives, the overall computational cost is dominated by two SPADE procedures and can therefore be expressed as

$$T_{\text{DPM}} = O((|\mathcal{U}| + N_{\text{pat}}) \cdot \bar{S}), \quad S_{\text{DPM}} = O(N_{\text{pat}} \cdot |\mathcal{U}| \cdot \bar{S}). \quad (23)$$

C.2 Contextualized Action Semantic Encoding.

The computational cost of CASE follows exactly from its two-stage procedure. First, constructing the action-level Markov transitions requires a single pass over all user behavior sequences to count one-step action transitions. The transition-construction stage costs $O(|\mathcal{U}| \cdot \bar{S})$.

Second, the context-guided sorting relies on two Markov-based structures. The cluster-level transition matrix $M_g \in \mathbb{R}^{m \times m}$ contains $m \ll |\mathcal{A}|$ semantic clusters, so extracting cluster transition chains introduces only negligible $O(m)$ overhead. The action-level transition matrix $M_l \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{A}|}$ has $|\mathcal{A}|^2$ possible entries in the worst case, leading to an upper-bound aggregation cost of $O(|\mathcal{A}|^2)$. After context aggregation, actions inside each cluster are ranked according to their transition scores, which costs $O(|\mathcal{A}| \log |\mathcal{A}|)$.

Combining all components, the overall computational complexity of CASE is

$$T_{\text{CASE}} = O(|\mathcal{U}| \cdot \bar{S} + |\mathcal{A}|^2 + |\mathcal{A}| \log |\mathcal{A}|), \quad (24)$$

while the space complexity is dominated by the storage of transition matrices:

$$S_{\text{CASE}} = O(|\mathcal{U}| \cdot \bar{S} + |\mathcal{A}|^2). \quad (25)$$

C.3 Pattern-Aware Information Fusion Model.

The fusion stage integrates PS-BERT and PD-GNN without altering their underlying asymptotic behavior. PS-BERT follows the standard BERT backbone [3], where the multi-head self-attention and projection modules incur a per-layer cost of $O(\bar{S}^2 \cdot d)$, with d denoting the hidden dimension. The pattern-aware masking introduces only element-wise token operations and an auxiliary loss of $O(\bar{S} \cdot d)$, which is negligible compared with the dominant quadratic term. The overall time complexity of PS-BERT thus remains $O(\bar{S}^2 \cdot d)$.

For space complexity, the main contributors are the attention score matrices of size $O(\bar{S}^2)$, the hidden states of size $O(\bar{S} \cdot d)$, and the trainable parameter matrices of size $O(d^2)$. Summing these terms yields a total space complexity requirement of $O(\bar{S}^2 + \bar{S} \cdot d + d^2)$.

PD-GNN augments a sparse GNN with an additional high-pass filtering branch. Both branches perform message passing over the same behavior graph, and the per-layer arithmetic cost is therefore dominated by the number of edges, giving $O(|\mathcal{E}| \cdot d)$. For space complexity, the model stores node representations of size $O(|\mathcal{V}| \cdot d)$, edge-level messages of size $O(|\mathcal{E}| \cdot d)$, and parameter matrices of size $O(d^2)$. The resulting space complexity is $O(|\mathcal{V}| \cdot d + |\mathcal{E}| \cdot d + d^2)$.

Combining the two components, the overall time and space complexity of the fusion model are as follows:

$$T_{\text{Fusion}} = O((\bar{S}^2 + |\mathcal{E}|) \cdot d), \quad (26)$$

$$S_{\text{Fusion}} = O(\bar{S}^2 + (\bar{S} + |\mathcal{V}| + |\mathcal{E}| + d) \cdot d). \quad (27)$$

C.4 Offline Preprocessing and Online Inference.

The first two modules, DPM and CASE, are executed entirely during offline preprocessing and therefore do not contribute to online latency.

Summing the three offline components yields the total offline complexity:

$$\begin{aligned} T_{\text{offline}} &= T_{\text{DPM}} + T_{\text{CASE}} + T_{\text{Fusion}} \\ &= O((|\mathcal{U}| + N_{\text{pat}}) \bar{S} + |\mathcal{A}|^2 + |\mathcal{A}| \log |\mathcal{A}| + (\bar{S}^2 + |\mathcal{E}|)d), \end{aligned} \quad (28)$$

$$\begin{aligned} S_{\text{offline}} &= S_{\text{DPM}} + S_{\text{CASE}} + S_{\text{Fusion}} \\ &= O(N_{\text{pat}} |\mathcal{U}| \bar{S} + |\mathcal{A}|^2 + \bar{S}^2 + (\bar{S} + |\mathcal{V}| + |\mathcal{E}| + d) \cdot d). \end{aligned} \quad (29)$$

During online inference, only the fusion model is invoked. Consequently, the time and space complexity of generating an online response are

$$T_{\text{online}} = O((\bar{S}^2 + |\mathcal{E}|) \cdot d), \quad (30)$$

$$S_{\text{online}} = O(\bar{S}^2 + (\bar{S} + |\mathcal{V}| + |\mathcal{E}| + d) \cdot d). \quad (31)$$

Because the computationally intensive components (DPM and CASE) operate solely offline, the online serving phase remains lightweight and efficient. In experiments, PIAD processes each batch in approximately 0.21s on an H20 GPU, demonstrating that its online inference is both efficient and practical for large-scale deployment.

D Examples of Mined Behavioral Patterns

In this section, we present examples of behavioral patterns mined from different datasets. Specifically, we illustrate one representative case from the Fraud dataset (romance scam accounts) and one from the Non-fraud dataset (pornographic dissemination accounts). Visual illustrations of the two cases are provided in Fig. 3a and Fig. 3b, respectively.

D.1 Romance Scam Accounts

Romance scam accounts are characterized by long-term identity fabrication and emotional manipulation. As shown in Fig. 3a, these accounts frequently modify their profile, like usernames and avatars, and often leverage QR-code-based links to redirect users off-platform, thereby obscuring their true identities and maintaining fabricated personas.

Correspondingly, their behavioral patterns typically exhibit a clear temporal structure. In the early stage, romance scam accounts demonstrate high-frequency profile modification behaviors and active friend-adding actions, aiming to increase exposure and establish multiple potential targets. In the later stage, their interactions become increasingly concentrated on several specific users, with sustained and high-frequency private messaging, reflecting prolonged engagement with the selected victims.

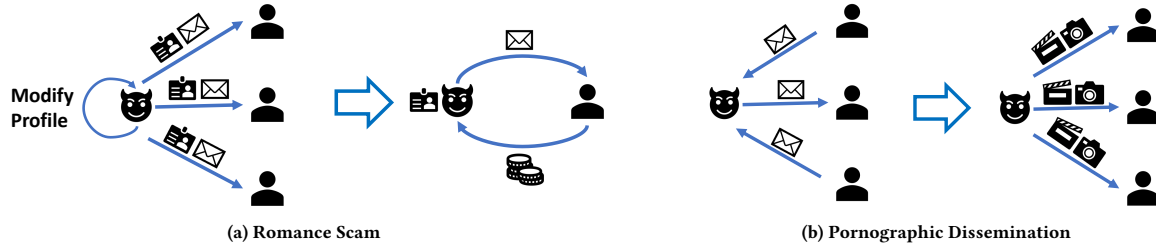


Figure 3: Illustrative examples of category-specific behavioral patterns, including Romance scam Accounts in the Fraud dataset and Pornographic dissemination Accounts in the Non-fraud dataset.

D.2 Pornographic Dissemination Accounts

In contrast, pornographic content dissemination accounts in the Non-fraud dataset follow a substantially different behavioral strategy. As illustrated in Fig. 3b, these accounts primarily attract users through sexually suggestive profile attributes such as avatars or personal descriptions.

Friendship establishment for such accounts is typically bidirectional, involving both active friend requests and passive acceptance from other users. Once a friend relationship is established, these

accounts rapidly transition into high-frequency content dissemination, sending messages that often contain explicit images or videos. Unlike romance scam accounts, their interactions emphasize short-term exposure and rapid content delivery rather than sustained one-to-one communication.

These two examples demonstrate that accounts from different datasets exhibit distinct and category-specific behavioral regularities. While romance scam accounts show a staged pattern with early identity manipulation followed by prolonged interaction with individual users, pornographic dissemination accounts emphasize rapid attraction and immediate high-volume content sharing.